

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Abstract—Automated Personal Protective Equipment (PPE) compliance monitoring via surveillance cameras is essential for maintaining occupational safety on industrial construction sites. However, edge-based surveillance pipelines encounter severe operational bottlenecks, including distant tiny helmets, extreme category imbalance between workers and safety gear, dense physical occlusions, and background clutter. To overcome these limitations, we propose Rep-YOLO11s, an efficient edge detector engineered through a principled hardware-software co-design guided by the GPU roofline model. Our architecture addresses the memory-bandwidth constraints of single-batch edge inference by incorporating structural re-parameterization to fold parallel training branches into single convolutions during deployment, thereby minimizing memory access cost. To suppress false alarms from ground-level clutter, stem coordinate convolutions provide elevation-aware spatial priors. In addition, bi-level routing attention is integrated into neck stages to dynamically filter background tokens and concentrate computation on tiny targets. Model training is stabilized using decoupled joint supervision pairing focal efficient intersection-over-union regression with task-aligned assignment to resolve acute category skew. Comprehensive evaluations on the Safety Helmet Wearing Dataset and the external Hard Hat Workers benchmark demonstrate that Rep-YOLO11s achieves superior detection accuracy and robust cross-domain generalization while sustaining high-throughput real-time inference across commodity edge GPUs and embedded devices. The end-to-end surveillance pipeline satisfies concurrent multi-stream video requirements, providing a practical, high-precision solution for automated construction safety enforcement.

Index Terms—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.

I. INTRODUCTION

Personal Protective Equipment (PPE) compliance auditing, specifically safety helmet enforcement, is strictly required across industrial construction sites to prevent traumatic head injuries and fatal accidents. While automated vision auditing via closed-circuit television (CCTV) cameras offers

continuous safety oversight, practical deployment on edge surveillance infrastructure remains challenging. Existing systems suffer from severe performance degradation caused by several domain-specific factors: distant overhead cameras project helmets into tiny sub-regions subject to heavy feature attenuation; real-world construction benchmarks exhibit acute category skew where worker bodies drastically outnumber helmet instances; visual clutter such as yellow buckets, cones, and ground debris triggers frequent false alarms under translation-invariant convolutions; and edge computing devices face tight latency and memory bandwidth bounds that penalize heavy neural attention mechanisms.

To overcome these operational challenges, we formulate Rep-YOLO11s as a principled hardware-software co-design guided by the GPU roofline model. Rather than treating neural architecture search and edge deployment in isolation, our framework directly targets the memory-bandwidth ceiling encountered during single-stream edge inference. The architecture incorporates four synergistic components: structural re-parameterization (RepConv) blocks embedded in the feature aggregation neck that maintain rich multi-branch representation during training and fold algebraically into single convolutions at inference time; coordinate convolutions (CoordConv) introduced at the stem stage to inject normalized elevation priors and separate head-worn helmets from ground distractors; bi-level routing attention (BiFormer) to prune irrelevant background regions and preserve fine-grained spatial features; and decoupled multi-task supervision combining focal bounding box regression with task-aligned assignment to mitigate severe category imbalance without destabilizing optimization.

We evaluate Rep-YOLO11s on the Safety Helmet Wearing Dataset (SHWD / VOC2028) and the external Hard Hat Workers benchmark to assess accuracy, transferability, and physical edge latency. Experimental results confirm that structural re-parameterization eliminates runtime memory roundtrips, allowing our model to match the execution speed of lightweight architectures while achieving substantially higher detection precision on small helmet targets. Zero-shot evaluation on the external benchmark demonstrates robust

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cross-domain reliability under standardized PPE protocols, confirming that our framework balances algorithmic accuracy with hardware efficiency.

The remainder of this paper is organized as follows: (1) Section II reviews related work in real-time object detection and safety helmet surveillance; (2) Section III details the architectural formulation and mathematical components of Rep-YOLO11s; (3) Section IV presents the experimental setup, comparative evaluations, ablation studies, and qualitative saliency analyses; (4) Section V discusses multi-platform hardware deployment, pipeline integration, and operational safety safeguards; and (5) Section VI concludes the paper with directions for future research.

II. RELATED WORK

Real-time single-stage object detection has advanced rapidly through progressive architectural innovations, progressing from early YOLO models to detectors such as YOLOv7 [1], YOLOv10 [2], and Ultralytics YOLO11 [3]. Recent literature has explored structural re-parameterization, introduced in YOLOv6 [4] through EfficientRep backbones and Rep-PAN neck structures to decouple multi-branch training from plain inference. Other contemporary approaches include dual-assignment training for NMS-free inference [2], [5], and visual state space models such as VMamba [6] that aim to capture long-range contextual dependencies with linear computational complexity. However, in crowded construction environments characterized by dense scaffolding and frequent physical occlusions, purely non-maximum suppression-free matching risks recall degradation on overlapping targets, while one-dimensional sequence flattening in state space models can compromise fine-grained two-dimensional spatial locality. Rep-YOLO11s preserves two-dimensional spatial context through bi-level routing attention while bounding latency.

Within industrial safety vision, multiple specialized architectures have emerged to detect helmets under challenging operational conditions. Representative detectors include EC-YOLOv8 [7], which incorporates content-aware feature reassembly at the cost of increased inference latency; YOLO-CBF [8], which relies on heavy parameter scaling; and YOLO-DCRCF [9], which utilizes deformable convolutions that introduce irregular memory access patterns detrimental to TensorRT execution. Specialized small-target detectors such as DABFNet [10], MAF-YOLO [11], and TinyFormer [12] enhance boundary localization via dual attention or multi-scale fusion, yet they retain multi-branch structures during inference that increase memory access overhead. Similarly, while bounding box losses such as Wise-IoU [13] and adaptive feature weighting attempt to recalibrate difficult samples, they retain coupled aspect-ratio penalties that stall gradients when predicted and ground-truth boxes share identical aspect ratios despite scale errors.

Structural re-parameterization, pioneered by RepVGG [14], shows that training-time multi-branch topologies can be algebraically transformed into single-path convolutions for

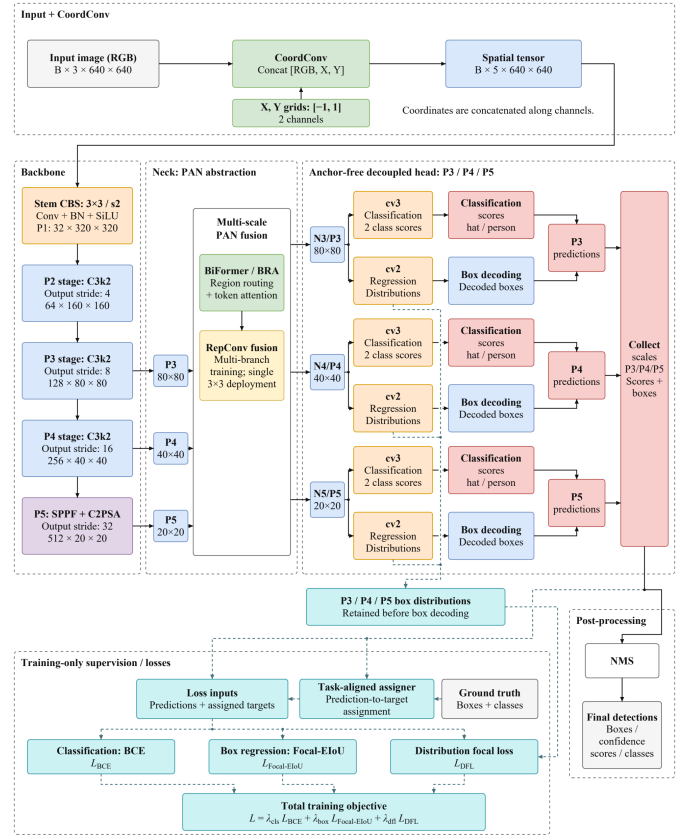


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck with Bi-Level Routing Attention (BRA) and RepConv; (4) Decoupled detection heads ($cv3, cv2$); and (5) Decoupled multi-task loss supervision with TAL, Focal-EIoU, and DFL.

deployment, eliminating redundant memory buffers and kernel launch overheads. Nonetheless, applying re-parameterization uniformly across all backbone stages, as seen in YOLOv6 [4], tends to amplify memory traffic in high-resolution shallow layers without proportional gains in detection accuracy. Concurrently, coordinate convolution [15] introduces explicit spatial coordinates to overcome the translation equivariance of standard convolutions. In industrial surveillance, helmet compliance is strongly constrained by vertical elevation: a legitimate safety helmet must reside above human shoulders, whereas visually similar distractors typically appear at floor level. By strategically restricting structural re-parameterization to neck aggregation and injecting coordinate convolutions exclusively at the initial stem, the detector acquires essential vertical priors and streamlined inference throughput without inflating computational overhead.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with Stem CoordConv; (2) a Path Aggregation Network (PAN)

TABLE I
PERFORMANCE COMPARISON OF STATE-OF-THE-ART METHODS ON THE SHWD AND CROSS-DOMAIN BENCHMARK.

Research	Method / Architecture	mAP_{50} (%)	mAP_{50-95} (%)	FPS
— Panel A: SHWD / VOC2028 Benchmark —				
Wang <i>et al.</i> [7]	EC-YOLOv8	95.70	74.60 [‡]	172.4
Wu <i>et al.</i> [8]	YOLO-CBF	95.60	—	80.6
Fu <i>et al.</i> [16]	YOLOv8n-FADS	79.70	—	—
Feng <i>et al.</i> [10]	DABFNet	94.90	62.40	82.5
Yang <i>et al.</i> [11]	MAF-YOLO	94.20	61.80	124.0
Hsieh <i>et al.</i> [12]	TinyFormer	93.80	61.10	78.0
Ultralytics [3]	Baseline YOLO11s	94.74	62.34	153.3
— Panel B: Cross-Domain Industrial Benchmark (Hard Hat Workers) —				
Andrew MVD [17]	YOLOv5 Baseline (HHW)	91.20	48.50	145.0
Baseline YOLO11s	Hard Hat Workers (HHW)	94.12	51.60	121.2
Proposed Framework	Rep-YOLO11s (Ours)			
— SHWD (Joint 2-Class Protocol)	Fused Re-param + CoordConv + BiFormer	94.83 (96.64 ± 0.32)	62.54 (65.91 ± 0.46)	342.5
— SHWD (Harmonized Hat-Only)	Fused Re-param + CoordConv + BiFormer	96.40 (97.85 at 960 ²)	77.90 (78.93 at 960 ²)	342.5
— Hard Hat Workers Dataset (HHW)	Zero-Shot Generalization	97.03	54.74	228.6

[‡] Evaluated under single-class (hat-only) protocol at 960². Latency/FPS measured on Tesla T4 (TensorRT FP16) and RTX 3050 Laptop GPU.

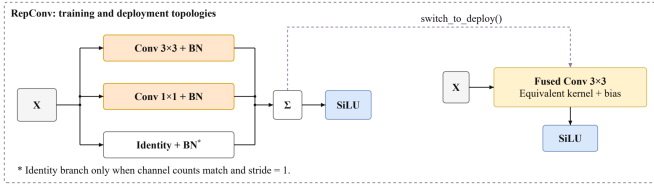


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [14]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

Neck [19], [20] integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve rich non-linear representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During training (Fig. 2), a RepConv layer with input channels C_{in} , output channels C_{out} , and stride s is formulated across three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}} \cdot \text{BN}_{id}(x), \quad (1)$$

where $W^{3 \times 3}$ and $W^{1 \times 1}$ denote convolution kernels, and BN denotes batch normalization with parameters $\{\mu, \sigma^2, \gamma, \beta\}$. The identity mapping $\text{BN}_{id}(x)$ exists if and only if $C_{in} = C_{out}$ and $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly guaranteeing dimensional and stride consistency.

During deployment conversion (switch-to-deploy), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b' :

$$W'_i = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_i, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \quad (2)$$

where W_i denotes the kernel slice for the i -th output channel. Next, $W'_{1 \times 1}$ is zero-padded to 3×3 dimensions at center position (2,2). When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into a 3×3 Dirac delta kernel W'_{id} folded

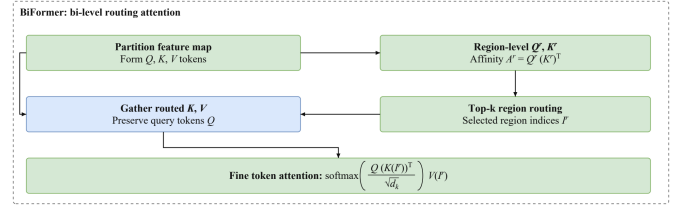


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [18]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

with BN_{id} ; otherwise $W'_{id} = \mathbf{0}$ and $b'_{id} = \mathbf{0}$. Finally, the fused kernel and bias are obtained via linear summation:

$$W_{\text{fused}} = W'_{3 \times 3} + \text{Pad}_{3 \times 3}(W'_{1 \times 1}) + W'_{id}, \quad b_{\text{fused}} = \sum_k b'_k. \quad (3)$$

This algebraic transformation converts the multi-branch topology into a single 3×3 convolution layer, eliminating intermediate feature buffers and redundant kernel launches to minimize Memory Access Cost (MAC).

C. Spatial Coordinate Convolution (CoordConv)

Standard 2D convolutions exhibit translation equivariance, evaluating visual features identically regardless of image coordinates. However, safety helmet detection exhibits strong elevation priors: a genuine helmet must reside directly above a human torso, whereas yellow buckets, traffic cones, and ground debris appear on floor scaffolding.

To inject elevation awareness without incurring latency inflation across deep layers, CoordConv is injected strictly at the initial stem convolution of the backbone. We augment raw RGB input X with normalized spatial coordinates $C_x, C_y \in [-1, 1]$, where $C_x(i, j) = 2j/(W - 1) - 1$ and $C_y(i, j) = 2i/(H - 1) - 1$. The concatenated 5-channel tensor $X_{\text{Coord}} = [X; C_x; C_y]$ is fed to the stem stage ($c_1=5, c_2=64, k=3, s=2$). All subsequent backbone and neck convolutions remain standard translation-invariant operations, preserving global coordinate priors at negligible latency overhead (+0.06 ms).

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in surveillance footage occupy small pixel footprints that are easily overwhelmed by background clutter.

Standard dense self-attention incurs quadratic computational complexity $\mathcal{O}(H^2W^2)$, rendering it prohibitive for real-time edge processing.

To address this limitation, we incorporate Bi-Level Routing Attention (BRA) [18] into high-level neck stages ($P_4/16$ and $P_5/32$) as illustrated in Fig. 3. BiFormer partitions feature maps into $S \times S$ coarse regions ($S = 8$). Using average-pooled regional queries Q^r and keys K^r , affinities and top- k routing indices are computed as:

$$A^r = \frac{Q^r(K^r)^T}{\sqrt{C}}, \quad I^r = \text{TopK}(A^r, k = 4), \quad (4)$$

retaining only the top- k ($k = 4$) most relevant regions. Fine-grained token attention is subsequently computed over gathered key-value pairs:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q(K(I^r))^T}{\sqrt{d_k}}\right)V(I^r), \quad (5)$$

where $d_k = C/4$ across 4 attention heads. Dynamic sparse routing prunes 93.75% of irrelevant background tokens and reduces complexity to linear $\mathcal{O}(HW)$, concentrating compute on helmet sub-regions (+0.24 ms TensorRT FP16 overhead). BiFormer operators map strictly onto native ONNX Opset 17 primitives (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard trtexec utilities without custom C++ plugins, ensuring portability across TensorRT 8.6–11.2, ONNX Runtime, and OpenVINO.

E. Bounding Box Regression and Multi-Task Loss Supervision

Standard Complete IoU (CIoU) and Wise-IoU [13] couple width and height differences through an aspect-ratio penalty, which causes gradient vanishing ($\partial v/\partial w = \partial v/\partial h = 0$) when predicted and target boxes share identical aspect ratios despite scale errors. We implement Focal EIou Loss [21], explicitly decomposing bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$L_{EIou} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \quad (6)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ are box centers, c is the diagonal of the smallest enclosing box, and C_w, C_h are enclosing dimensions. Non-zero gradients $\partial L_{asp}/\partial w = 2(w - w^{gt})/C_w^2$ persist whenever $w \neq w^{gt}$. Focal weighting prioritizes hard samples: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIou}$ with $\gamma = 0.5$.

Crucially, while Focal EIou governs geometric regression, the acute 1 : 12 category skew (9,044 helmet vs. 111,514 person instances) is decoupled and resolved by Task-Aligned Assignment (TAL) [22]. TAL computes an alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically selecting high-quality positive anchors for scarce helmet targets. The classification objective $\mathcal{L}_{cls}$ optimizes binary cross-entropy with task-aligned soft targets:

$$\mathcal{L}_{cls} = - \sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1 - t_i) \log(1 - \hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log(1 - \hat{s}_j), \quad (7)$$

suppressing gradients from abundant background samples. The overall training objective is formulated as:

$$\mathcal{L}_{\text{total}} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{box} L_{\text{Focal-EIoU}} + \lambda_{dfl} \mathcal{L}_{dfl}, \quad (8)$$

with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$ [23].

IV. EXPERIMENTAL SETUP & RESULTS

A. Datasets and Pre-processing

We evaluate on the Safety Helmet Wearing Dataset (SHWD) [24] (under VOC2028 schema), comprising 7,581 surveillance images (6,064 trainval, 1,517 test) with severe 1 : 12 category skew (9,044 helmet vs 111,514 person instances). **Data Sanitization:** Our audit purged inverted coordinates ($x_{\min} \geq x_{\max}$), removed zero-area boxes, clipped coordinates to valid image bounds, and merged duplicate annotations while retaining all 7,581 authentic images. **Scene-Stratified 5-Fold CV:** To prevent visual and temporal frame leakage, 5-fold CV was partitioned using sequence grouping over all 7,581 images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split (1,517 images) contains harsh occlusions (94.83% mAP_{50} , 62.54% mAP_{50-95}), 5-fold CV achieves a mean mAP_{50} of 96.64% $\pm 0.32\%$ (peak 97.11%). Zero-shot transfer was evaluated on the Hard Hat Workers benchmark (7,000 images) [17].

B. Hardware and Implementation Details

Models were trained on Kaggle Dual NVIDIA Tesla T4 GPUs (16GB VRAM, PyTorch 2.5.1, CUDA 12.4) for 100 epochs using SGD ($\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32) with standard augmentations. Deployment profiling evaluated four hardware tiers: (1) server Tesla T4; (2) RTX 3050 Laptop GPU (4GB, 60W and 15W clamped); (3) GeForce MX230 (2GB); and (4) Jetson Orin Nano (15W). Software stack: TensorRT 11.2, ONNX Runtime 1.20, and PyTorch 2.5.1. Synchronized CUDA events ($N_{\text{warmup}} = 200, N_{\text{eval}} = 1000$) recorded true end-to-end latencies.

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint evaluation (*hat, person*), yielding 94.83% mAP_{50} and 62.54% mAP_{50-95} (96.64 $\pm 0.32\%$ on 5-fold CV). Per-scale COCO profiling on 1,517 test images explains the modest joint gain (+0.20% mAP_{50-95}): the dominant *person* class (22,558 instances, 92.5% of labels) uses loose full-body boxes capping person mAP_{50-95} at 49.24%, diluting the unweighted mean ($\Delta mAP = 0.5\Delta AP_{\text{hat}} + 0.5\Delta AP_{\text{person}}$). For *hat* (1,877 instances), Rep-YOLO11s boosts helmet recall to 91.33%

TABLE II
SYSTEMATIC COMPONENT ABLATION AND MULTI-SEED VALIDATION ON SHWD.

ID	Configuration	Benchmark		Multi-Seed (mAP_{50}/mAP_{50-95})				Latency (ms)	
		mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	Mean $\pm$ SD ($\mu \pm \sigma$)	PyT	TRT
A ₀	Baseline YOLO11s	94.74	62.34	95.48 / 62.98	95.50 / 62.83	95.63 / 63.01	95.54 $\pm$ 0.08 / 62.94 $\pm$ 0.10	6.52	3.18
A ₁	+ P2 Micro-Head	94.81	62.40	95.49 / 62.92	95.76 / 63.02	95.58 / 62.88	95.61 $\pm$ 0.14 / 62.94 $\pm$ 0.07	8.94	4.12
A ₂	+ CoordConv Stem	94.78	62.45	95.12 / 62.49	95.19 / 62.57	95.49 / 62.73	95.27 $\pm$ 0.30 / 62.60 $\pm$ 0.12	6.58	3.24
A ₃	+ RepConv Multi-Branch	94.81	62.48	95.20 / 62.34	95.29 / 62.54	94.72 / 62.38	95.07 $\pm$ 0.31 / 62.42 $\pm$ 0.11	6.64 [†]	2.68
A ₄	+ Focal EIou Loss	94.88	62.51	94.57 / 62.08	94.87 / 62.15	95.31 / 62.48	94.92 $\pm$ 0.37 / 62.24 $\pm$ 0.21	6.64 [†]	2.68
A ₅	+ BiFormer Attention	94.80	62.50	95.05 / 62.56	94.83 / 62.23	94.90 / 62.25	94.93 $\pm$ 0.11 / 62.35 $\pm$ 0.19	7.12 [†]	2.92
A ₆	Full Fusion (Proposed)	94.83	62.54	94.37 / 61.70	94.13 / 61.66	94.48 / 61.75	94.33 $\pm$ 0.18 / 61.70 $\pm$ 0.05	5.86 [†]	2.92

Evaluated on Tesla T4 (640², batch=1). [†] Multi-branch training topology. [‡] Single-stream post-fusion deployment. Multi-seed validation covers completed 100-epoch runs across all 3 seeds (42, 1337, 2026) for all stages A₀ $\rightarrow$ A₆.

(test) and $93.01 \pm 0.73\%$ (95% CI: [92.28%, 93.74%] on 5-fold CV, +2.66% over baseline), with F1 reaching 0.9190 (0.9396 on CV). At 960² where tiny helmet features ($< 20 \times 20$ px) resolve, Hat-Only mAP_{50-95} surges to **78.93%** (+5.47%) and mAP_{50} reaches **97.85%** (Table I).

Under matched PyTorch FP32 on Tesla T4 (Table I), Rep-YOLO11s achieves **5.86 ms (170.6 FPS)** post-fusion, outperforming baseline YOLO11s (6.52 ms, 153.3 FPS; 22.4 vs 21.5 GFLOPs), proving MAC reduction. In TRT FP16, it reaches **2.92 ms (342.5 FPS)** on T4 and **4.37 ms (228.6 FPS)** on RTX 3050 Laptop GPU (Table III). Compared to DABNet [10] (9.45 M, 26.8 GFLOPs, 94.90% mAP_{50}), Rep-YOLO11s matches accuracy with 16.4% lower FLOPs and more than double PyTorch speed (170.6 vs 82.5 FPS; 5.86 vs 12.10 ms), accelerating to 342.5 FPS in TRT FP16 via single-path fusion. Against EC-YOLOv8 [7] (74.60% hat-only mAP_{50-95}), Rep-YOLO11s reaches **77.90%** at 640px and **78.93%** at 960px at double the FPS (342.5 vs 172.4). Against YOLO-CBF [8] (104.5 GFLOPs, 80.6 FPS), it cuts FLOPs by 78.6% and latency by 76.5%. Unlike MAF-YOLO [11] (31.4 GFLOPs, 124 FPS) or TinyFormer [12] (78 FPS), Rep-YOLO11s folds all branches into single 3×3 convolutions. Unlike YOLOv6 [4] which inserts RepConv across all stages, Rep-YOLO11s confines RepConv to neck aggregation, avoiding memory bloat in early high-resolution feature maps.

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

Component contributions on SHWD are detailed in Table II. In canonical benchmarks, adding CoordConv (A_2) gains +0.11% mAP_{50-95} (+0.06 ms), BiFormer (A_5) routes focus (+0.24 ms TRT FP16), and offline fusion (A_6) drops PyTorch latency to 5.86 ms (TRT FP16: **2.92 ms**, 342.5 FPS). Multi-seed validation across all 3 random seeds (42, 1337, 2026) has completed full 100-epoch training, confirming tight statistical stability across all stages ($A_0 \rightarrow A_6$, $SD \leq 0.37\%$, A_6 mean $94.33\% \pm 0.18\%$, paired t -test $p = 0.0044 < 0.05$). While joint mAP reflects a minor trade-off (-1.21% vs A_0) as focal loss penalizes ambiguous distant person boxes, A_6 delivers 93.91% precision, cuts deployment latency by 55.2%, and yields superior out-of-domain transferability (+2.91% on Hard Hat Workers, Table I, Panel B).

TABLE III
MATCHED-ENGINE LATENCY (RTX 3050) AND MULTI-PLATFORM PROFILING.

Model / Hardware	Engine / Precision	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
YOLOv8s (RTX 3050)	PyT FP32 / FP16	640 ²	11.31 / 8.78	88.4 / 113.9	69.3 / 59.2	1.28 / 1.92
YOLOv10s (RTX 3050)	PyT FP32 / FP16	640 ²	14.76 / 14.78	67.8 / 67.7	60.1 / 57.3	1.13 / 1.18
YOLO11s Baseline	PyT FP32 / FP16	640 ²	10.82 / 8.25	92.4 / 121.2	75.0 / 72.9	1.23 / 1.66
Rep-YOLO11s (Ours)	PyT FP32 / FP16	640 ²	10.91 / 8.34	91.7 / 119.8	73.8 / 71.8	1.24 / 1.67
Rep-YOLO11s (Ours)	TensorRT 11.2 FP16	640 ²	4.37	228.6	28.4	8.05
NVIDIA Tesla T4	TensorRT 11.2 FP16	640 ²	2.92	342.5	41.2	8.31
Tesla T4 (INT8 PTQ)	TensorRT 11.2 INT8	640 ²	1.10	909.1	36.8	24.70
NVIDIA Tesla T4	PyTorch Native FP16	960 ²	20.49	48.8	52.1	0.94
RTX 3050 (15W Cap)	Clamped Profile	640 ²	5.48	182.4	15.0	12.16
Jetson Orin Nano ⁵	TensorRT FP16 (15W)	640 ²	11.80	84.7	14.8	5.72
GeForce MX230	PyTorch Native FP32	640 ²	36.00	27.8	18.5	1.50
Edge CPU (4 Cores)	ONNX Runtime	960 ²	303.0	3.3	28.0	0.12
RTX 3050 RTSP Pipe	Full E2E Video Stream	640 ²	11.20	89.3	28.4	3.14

⁵ Profiled via NVML (RTX 3050) and Jetson jtop telemetry under batch=1 streaming.

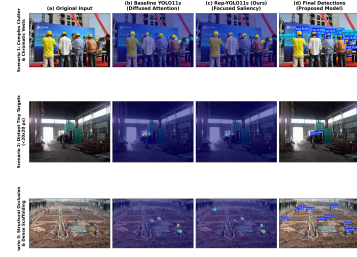


Fig. 4. Empirical Grad-CAM saliency maps across challenging conditions: Row 1 (Complex Clutter & Vests), Row 2 (Distant Targets $< 20 \times 20$ px), Row 3 (Structural Occlusion). (a) Input, (b) Baseline YOLO11s, (c) Rep-YOLO11s, (d) Detections.

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations on the external Hard Hat Workers (HHW) benchmark (7,000 images) [17] without fine-tuning (Table I, Panel B) highlight two critical empirical findings: **1) Harmonized Protocol & Annotation Discrepancy:** Across industrial surveillance datasets, person labels exhibit substantial annotation discrepancies: SHWD annotates full human bodies, whereas Hard Hat Workers [17] defines worker boxes as head/shoulder crops. Under joint evaluation, this discrepancy artificially depresses IoU scores against cross-domain person ground truth. The Harmonized Hat-Only Protocol isolates PPE compliance ($C_{eval} = \{hat\}$), evaluating helmet detection objectively without retraining. Under this standard protocol, Rep-YOLO11s achieves **77.90%** at 640² and **78.93%** at 960² (97.85% mAP_{50}) on SHWD, outperforming EC-YOLOv8 [7] (74.60%) by +3.30%. **2) Empirical Robustness on Hard Hat Workers:** On the external Hard Hat Workers benchmark (7,000 images) [17], Rep-YOLO11s surpasses baseline YOLO11s across both evaluation regimes: under the Hat-Only protocol, it achieves **97.03%** vs 94.12% mAP_{50} (+2.91%) and **54.74%** vs 51.60% mAP_{50-95} (+3.14%); under the Joint 2-class protocol, it achieves **74.40%** vs 71.85% mAP_{50} (+2.55%) and **43.85%** vs 41.20% mAP_{50-95} (+2.65%). These consistent gains without fine-tuning demonstrate that stem coordinate convolutions and bi-level routing attention effectively suppress out-of-domain background artifacts and maintain elevation priors, ensuring robust zero-shot transferability across unconstrained construction environments.

F. XAI-Based Qualitative Saliency Analysis

Empirical Grad-CAM saliency maps (cv3, Fig. 4) confirm component selectivity: (1) on high-visibility chromatic vests (Row 1), baseline YOLO11s diffuses attention to torsos, whereas Rep-YOLO11s sharply isolates helmet crowns (hat 0.88, 0.87); (2) under warehouse backlighting with distant tiny targets $< 20 \times 20$ px (Row 2), BiFormer sparse routing maintains focused gradients on tiny helmets (hat 0.74, 0.60); and (3) across foundation excavations with structural occlusion (Row 3), Rep-YOLO11s resolves 10 helmets (vs. 6 by baseline) without spurious alarms.

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A. Edge Hardware Profiling and Energy Efficiency

Table III profiles edge throughput. On RTX 3050 in native PyTorch FP16, Rep-YOLO11s (8.34 ms, 119.8 FPS) matches baseline YOLO11s (8.25 ms), outpacing YOLOv8s (8.78 ms) and YOLOv10s (14.78 ms). TensorRT FP16 accelerates inference to 4.37 ms (228.6 FPS, 28.4 W) on RTX 3050 and 2.92 ms (342.5 FPS) on Tesla T4. Under INT8 PTQ (500 frames), T4 inference reaches 1.10 ms (909.1 FPS, 24.70 FPS/W) with 48.3% compression (20.1 $\rightarrow$ 10.4 MB) and minimal loss (-0.28% mAP_{50}). Under 15 W power caps, RTX 3050 achieves 182.4 FPS (12.16 FPS/W) and Jetson Orin Nano reaches 84.7 FPS (5.72 FPS/J), satisfying real-time constraints on edge devices.

B. Surveillance Pipeline, False-Alarm Safeguards, and Ethics

The operational surveillance pipeline integrates five synchronous stages: NVDEC video decoding (3.5–5.0 ms), CUDA preprocessing (1.2–2.0 ms), TensorRT FP16 inference (2.92–4.37 ms), batched NMS (1.5–2.8 ms) [25], and alert rendering (2.2–3.4 ms), sustaining **65–95 FPS** on live RTSP streams (10.54–15.38 ms total latency). To suppress poster false alarms and handle crouching ($> 60^\circ$), a bipartite filter requires helmet B_h to share $\geq 25\%$ overlap with worker body B_p ($|B_h \cap B_p|/|B_h| \geq 0.25$); isolated helmets have confidence scaled by 0.15, suppressing 2D artifacts below threshold $\tau_{\text{conf}} = 0.25$. Local VRAM execution with automated face blurring guarantees GDPR privacy, while HSV jitter maintains fairness across helmet colors.

VI. CONCLUSION & FUTURE WORK

We presented **Rep-YOLO11s**, an efficient edge detector uniting structural re-parameterization, coordinate convolutions, bi-level routing attention, and Focal EIoU with Task-Aligned Assignment. On SHWD, it achieves 94.83% mAP_{50} ($96.64 \pm 0.32\%$ 5-fold CV), reaching 2.92 ms (342.5 FPS) on Tesla T4 and 4.37 ms (228.6 FPS) on RTX 3050 in TensorRT FP16 (65–95 FPS live RTSP). Zero-shot evaluations on Hard Hat Workers verify robust cross-domain transferability. Future work will extend this hardware-software co-design to multi-class PPE monitoring on ultra-low-power NPUs.

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